

ASSESSMENT TOOL - 3

Descriptive written test

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Artificial Intelligence

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1. Intelligent Agents in a smart home automation system

i) simple reflex agent

Definition:-

A simple reflex agent makes decisions based only on the current sensor input using predefined IF-THEN rules. It does not store past information or learn from experience.

Function in the smart home:-

- * If motion is detected $\rightarrow$ turn on the lights.
- * If room temperature exceeds 28°C $\rightarrow$ turn on the air conditioner.
- * If a door opens while the alarm is armed $\rightarrow$ Activate the security alarm.

Advantage:-

- * Fast response to real time events.
- * Easy to design and implement
- * Low computational cost.

Limitations

- * cannot remember previous events
- * cannot adapt to changing user preferences
- * poor performance in complex situations

2) model-based agent:-

A model-based agent maintains an internal model of the environment and updates it using current and past sensor information.

Function in the smart home

- * Remembers whether people are inside the house.
- * Keeps lights on even if no movement is detected for a short time.
- * Tracks previous temperature readings to avoid frequent AC switching.
- * Monitors whether doors and windows are locked.

Advantages:-

- * Handles partially observable environments.
- * Makes more accurate decisions using memory.
- * Improves system reliability.

Limitations:-

- * More complex than a simple reflex agent.
- * Requires additional memory and processing.

3) Goal-based agent:-

A goal-based agent selects actions that help achieve specific objectives.

Function in the smart home

- * Maintain room temperature at 24°C .
- * Reduce electricity consumption.
- * Keep the home secure.
- * Ensure all lights are turned off when the house is empty.

Function

The agent evaluates different actions and chooses the one that best achieves the desired goal.

Advantages:-

- * Make intelligent long term decisions.
- * Suitable for planning multiple actions.

Limitation:-

- * Requires more computation
- * Does not directly compare different levels of satisfaction.

4) Utility Based Agent:

- * A utility based agent evaluates each possible action using utility.
- * Selects the action with the highest overall benefit.
- * Examples: Balances comfort, energy efficiency and security.
- * Pros: Gives the best overall performance.
- * Cons: Most complex.

Conclusion:-

A hybrid intelligent agent provides the best balance of comfort, energy efficiency and security in a smart home system.

2) Robot maze problem using uninformed search

A robot is placed in an unknown maze and must find the exit without knowing the layout. It can use uninformed search algorithm such as uniform cost search and iterative deepening search to explore the maze.

Uniform cost search (UCS)

UCS expands the path with the lowest cumulative cost first. If different paths have different movement costs, UCS guarantees the optimal path to the exit.

Advantages:

- * Finds the optimal solution
- * Works well when path costs vary.

Disadvantages:

- * High time
- * High space

$$\text{Complexity: } O(b^{(1+LC/\epsilon)})^*$$
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Iterative Deepening
IDS repeatedly performs depth-limited search with increasing depth limits until the goal is found. It combines the benefits of depth-first search and breadth-first search.

Advantages:

- * complete and optimal for equal step costs.
- * low memory usage.
- * Space complexity: $O(bd)$

DPS advantages:

- * Repeats searches at each depth
- * Time complexity: $O(b^d)$

Intelligent agents:

- * simple reflex agent: Reacts only to current surroundings. unsuitable for an unknown maze.
- * Model-Based Agent: maintains memory of explored paths and visited locations. preventing repeated
- * Goal-based agent: searches for actions that achieve the goal of reaching the exit.
- * Utility-Based Agent: Chooses paths that maximize efficiency by considering distance, cost and safety

Conclusion:

For an unknown maze, UCS is best for finding the optimal path, while IDS is preferred when memory is limited. A model-based Goal-Based Agent combined with UCS provides the most reliable and intelligent solution for efficient maze navigation.